

Topic 2: Genes and health

This topic considers the following biological principles: the properties of and transport of materials, across cell membranes and gas exchange surfaces, DNA structure and replication, protein synthesis, enzymes and monohybrid inheritance through the context of the genetic disease cystic fibrosis. The potential that gene therapy offers as treatment for cystic fibrosis is examined. The topic also allows for discussion of the social and ethical issues surrounding the genetic screening for genetic conditions.

5.2 Assessment information

This unit is assessed by means of a written examination paper, which carries 80 marks, lasts 1 hour 30 minutes and will include:

- objective questions
- structured questions
- short-answer questions

and will also cover:

- *How Science Works*
- practical-related questions.

Students may use a calculator.

The quality of written communication will be assessed in the context of this unit through questions which are labelled with an asterisk (*). Students should take particular care with spelling, punctuation and grammar, as well as the clarity of expression, on these questions.